Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications

⚙️ Metadata

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📜 Abstract

Amid growing efforts to leverage advances in large language models (LLMs) and visionlanguage models (VLMs) for robotics, Vision-Language-Action (VLA) models have recently gained significant attention. By unifying vision, language, and action data at scale, which have traditionally been studied separately, VLA models aim to learn policies that generalise across diverse tasks, objects, embodiments, and environments. This generalisation capability is expected to enable robots to solve novel downstream tasks with minimal or no additional task-specific data, facilitating more flexible and scalable real-world deployment. Unlike previous surveys that focus

narrowly on action representations or high-level model architectures, this work offers a comprehensive, full-stack review, integrating both software and hardware components of VLA systems. In particular, this paper provides a systematic review of VLAs, covering their strategy and architectural transition, architectures and building blocks, modality-specific processing techniques, and learning paradigms. In addition, to support the deployment of VLAs in real-world robotic applications, we also review commonly used robot platforms, data collection strategies, publicly available datasets, data augmentation methods, and evaluation benchmarks. Throughout this comprehensive survey, this paper aims to offer practical guidance for the robotics community in applying VLAs to real-world robotic systems. All references categorized by training approach, evaluation method, modality, and dataset are available in the table on our project website: https://vla-survey.github.io.

📚 Zotero links

- Local library

Notes

🧐 Personal Summary

Design Transition :

Early CNN-based End-to-End - CLIPort

Transformer-based Sequence Models - Gato, VIMA

Unified real world policies with pre-trained VLMs - RT-1/2, RT - X series -> OpenVLA

Diffusion Policy - Octo (Only at action head)

Diffusion Transformer Architecture - RDT-1B

Flow Matching Policy Architectures - pi0 - action

Latent Action - LAPA (unlabeled video data for pre-training) , VQ-VAE (Xt to Xt+H)

Hierarchical Policy Architectures - Pi 0.5, GR00T N1

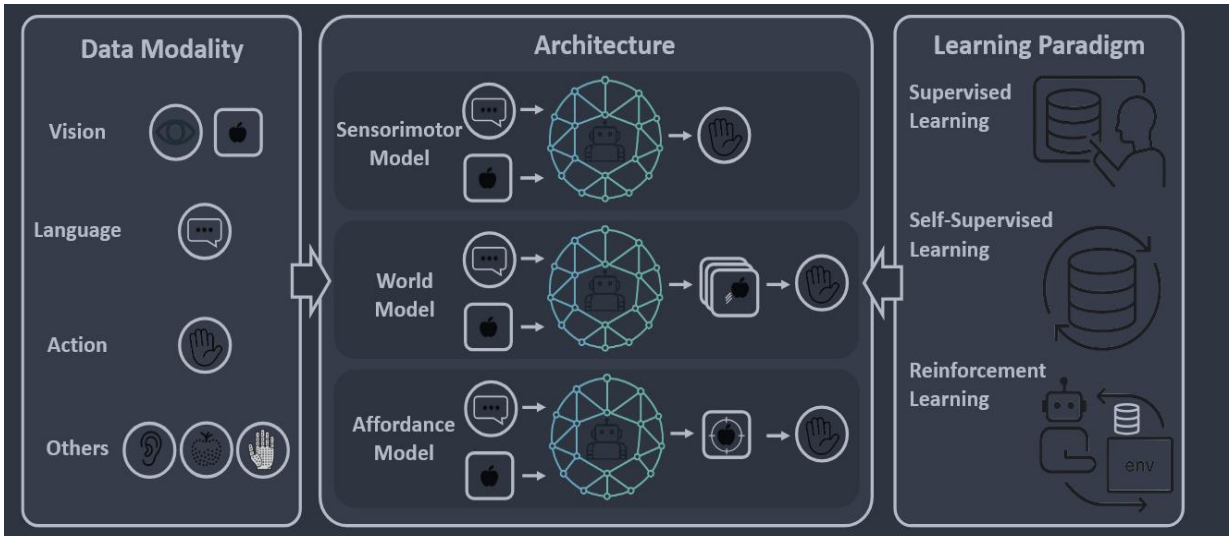
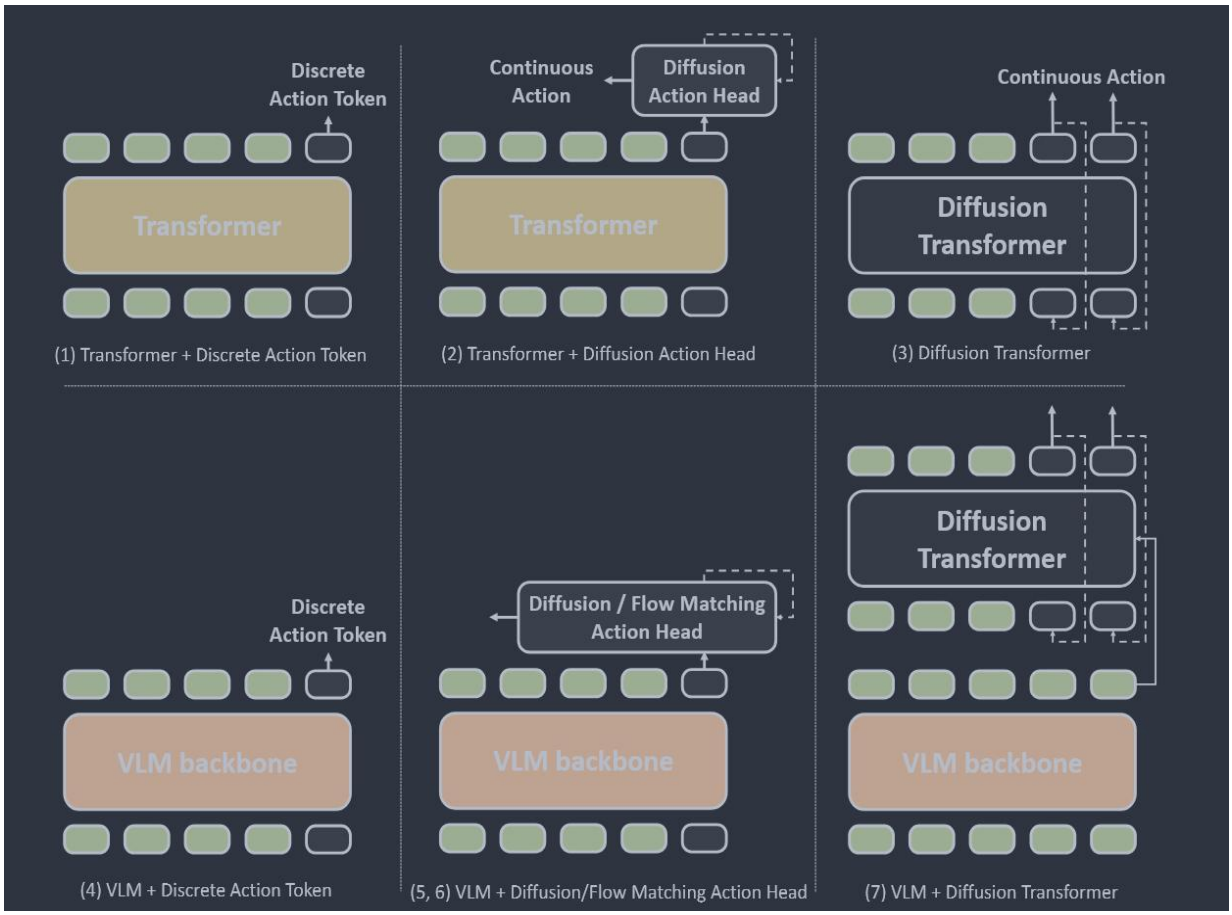


FIGURE 3. Structure of Section IV and Section V. The figure summarizes key components of VLA models. The center illustrates core architectural types, including sensorimotor models, world models, and affordance-based models. The left side depicts the input and output modalities—vision, language, action, and other auxiliary modalities. The right side presents training strategies, including supervised learning, self-supervised learning, and reinforcement learning, along with practical implementation considerations.

Architectures and building blocks

1.Sensorimotor Model



(1) Transformer + Discrete Action Token - VIMA, Gato, RT-1

(2) Transformer + Diffusion Action Head - Octo, NoMAD, TinyVLA, RoboBERT, VidBot

(3) Diffusion Transformer - RDT-1B - Action tokens via cross attention with a vision and language query + MLP mapping

(4) VLM + Discrete Action Token - RT-2 (PaLM-E or PaLI-X), LEO, GR-1, RTH, RoboMamba, QUAR-VLA, OpenVLA, LLARA, ECOT, 3D-VLA

(5) VLM + Diffusion/Flow Matching Action Head - Diffusion VLA, DexVLA, ChatVLA, ObjectVLA, GO-1, PointVLA, MoLe-VLA, Fis-VLA, CronusVLA

(6) VLM + Flow Matching Action Head - pi0, OneTwoVLA, Hume, SwitchVLA

(7) VLM + Diffusion Transformer - GR00T N1

2.World Model

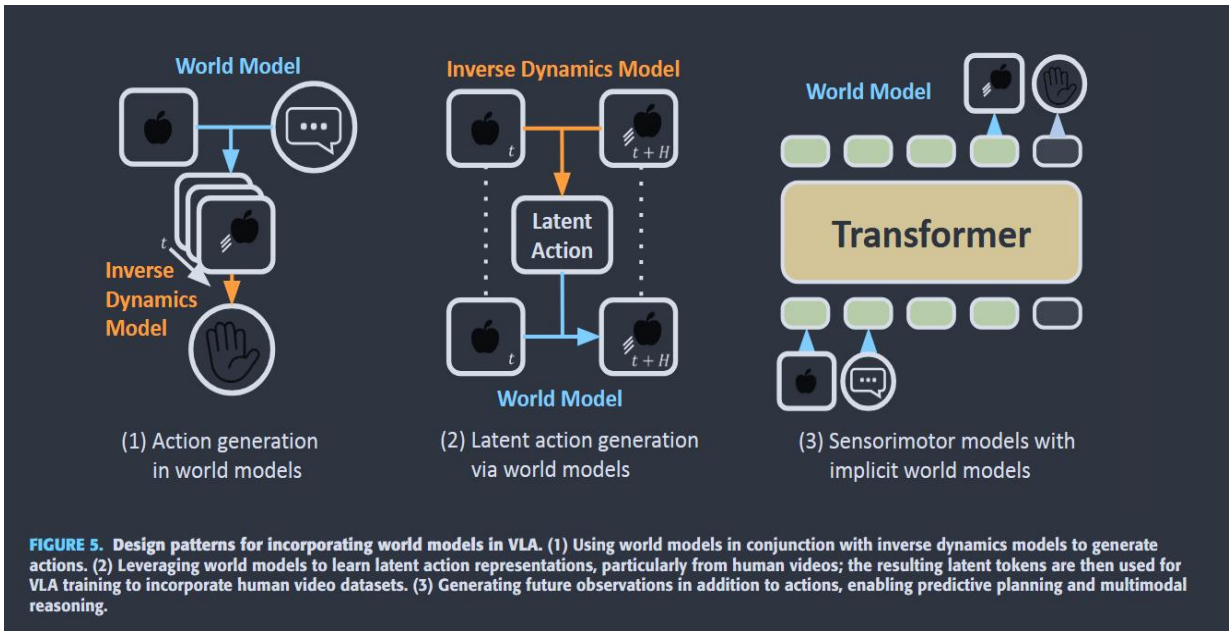


FIGURE 5. Design patterns for incorporating world models in VLA. (1) Using world models in conjunction with inverse dynamics models to generate actions. (2) Leveraging world models to learn latent action representations, particularly from human videos; the resulting latent tokens are then used for VLA training to incorporate human video datasets. (3) Generating future observations in addition to actions, enabling predictive planning and multimodal reasoning.

(1) Action generation - Future Observations - UniPi, DreamGen, GeVRM(Future visual representations) - IDM

LUMOS, CoT-VLA, MegaPose, Dreamitate

Optical Flow/Feature Point - AVDC, ATM, Track2Act, LangToMo, DiffMatcher, PPI

(2) Latent Action Generation - LAPA (Video -> latent action -> action head -> action)

(3) Sensorimotor models with implicit world models

jointly output action and future observations - GR-1, GR -2 GR-MG, WorldVLA, ViSA-Flow, FLARE, UVA

3. Affordance-Based Model

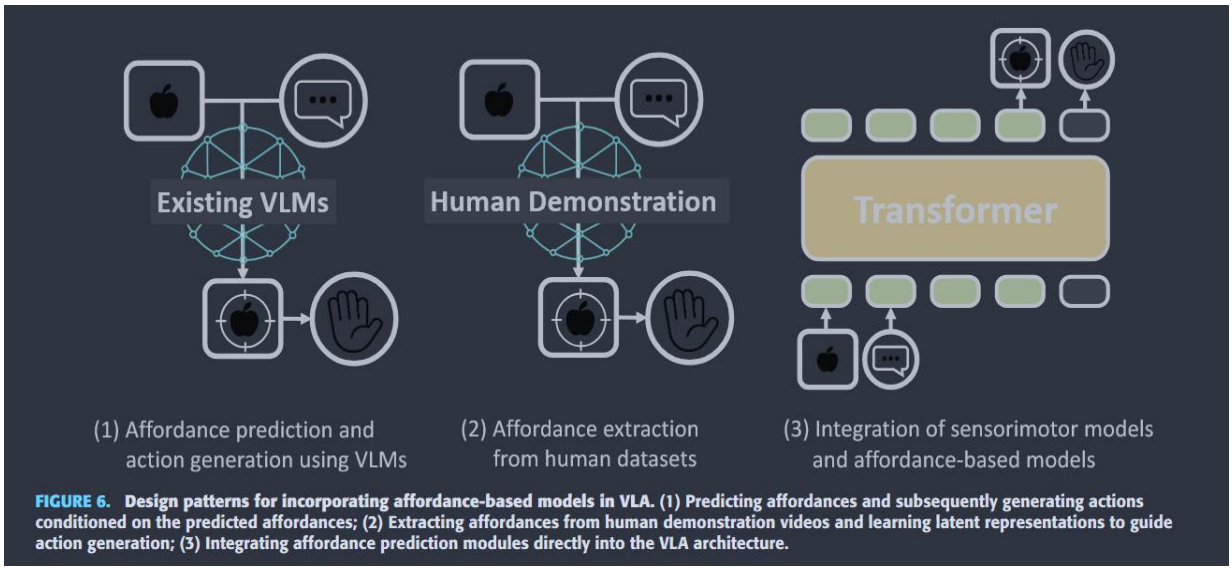


FIGURE 6. Design patterns for incorporating affordance-based models in VLA. (1) Predicting affordances and subsequently generating actions conditioned on the predicted affordances; (2) Extracting affordances from human demonstration videos and learning latent representations to guide action generation; (3) Integrating affordance prediction modules directly into the VLA architecture.

DATA MODALITIES

(1) Vision

most common: ResNet or ViT

VLM: CLIP, SigLip + DINOv2

Discretizing into token sequences: VQ-GAN, VQ-VAE

(2) Language Tokenization - Inherit from LLM - BPE, USE

(3) Action Representation:

a. Discretized action token obtained via binning

discretize each dimension of the action space into bins (typically 256), with each bin ID treated as a discrete token - RT-2, OpenVLA, FAST (DCT + BPE to reduce token length)

b. Decoding tokens into continuous actions

tokens generated by a transformer are mapped to continuous actions via a multilayer perceptron (MLP), typically trained with an L2 or L1 loss. - OpenVLA-OFT

c. Continuous action modeling via diffusion or flow matching - Octo, pi0

d. learning latent action representation from web-scale data - LAPA Moto UniVLA UniSkill

e. alternative action representation - SpatialVLA (allocate higher resolution to frequently occurring motions) ForceVLA, ChatVLA (MoE) switch action policies on task phases, iManip (Continual Learning)

f. Cross-embodiment action representation

OXE

CrossFormer - unified processing by first tokenizing heterogeneous sensor observation, unified token sequence + action head

UniAct - UAS

LangToMo, ATM -> Optical Flow

g. Miscellaneous Modalities

audio - Unified-IO 2, SOLAMI, FuSe, VLAS, MultiGen

tactile - FuSe TLA VTLA Tactile-VLA, ForceVLA

3D information - Depth images, Multi-view images, Voxel representations, Point clouds

Emerging Techniques:

Hierarchical Architectures - Atomic Skills, LMM-3DP

Chain of Thought Reasoning - ECoT, CoT-VLA

Training Strategies

Supervised Learning

Self-supervised Learning

- Modality Alignment
- Visual Representation
- Latent Action Representation Learning

Reinforcement Learning

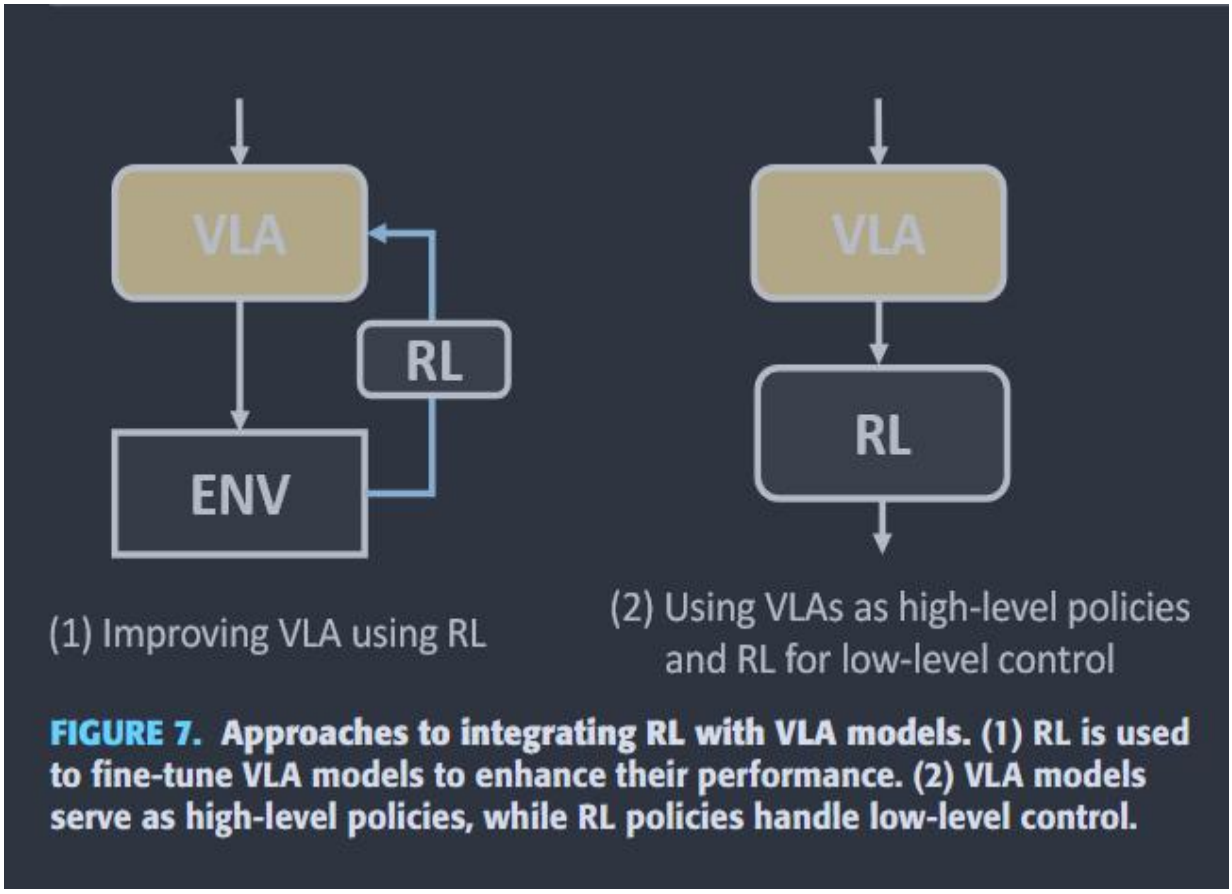


FIGURE 7. Approaches to integrating RL with VLA models. (1) RL is used to fine-tune VLA models to enhance their performance. (2) VLA models serve as high-level policies, while RL policies handle low-level control.

(1)iRe-VLA, VLA-RL, MoRE

(2)NaVILE, SLIM, LUMOS, DexTOG

Training Stages

a. Pre-training

b. Post-training - Fine-tune Action Head or LoRA or Distillation

Inference - RTC

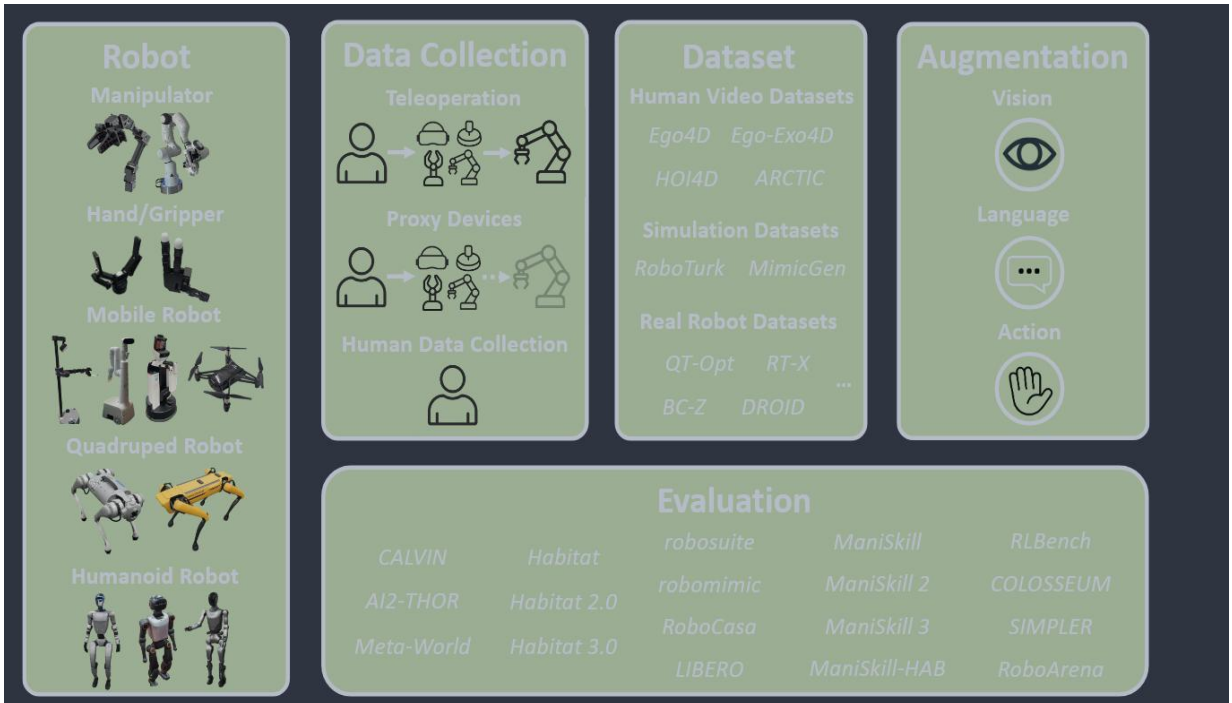


FIGURE 8. Structure of Section VI and Section VII: robots used VLA research — including manipulator, hand/gripper, mobile robot, quadruped robot, and humanoid robot; data collection methods — including teleoperation, proxy devices, and human data collection; publicly available dataset — including human egocentric data, simulation data, and real-world robot data; augmentation for vision, language, and action; various evaluation benchmarks.

TABLE 1. Recent real-world robot datasets used in VLA research. Here, *Skill* denote atomic action primitives (e.g., pick, place, reach), whereas *Task* correspond to instruction-level goals. All statistics are reported as in the original papers; the table is adapted from prior works [11], [330], [333], [335].

Name	Episodes	Skill	Task	Modality	Embodiment	Collection
QT-Opt [336]	580K	1 (Pick)	NA	RGB	KUKA LBR iiwa	Learned
MT-Opt [337]	800K	2	12	RGB, L	7 robots	Scripted, Learned
RoboNet [338]	162K	NA	NA	RGB	7 robots	Scripted
BridgeData [339]	7.2K	4	71	RGB, L	WidowX 250	Teleop
BridgeData V2 [340]	60.1K	13	NA	RGB-D, L	WidowX 250	Teleop
BC-Z [341]	26.0K	3	100	RGB, L	Google EDR	Teleop
Language Table [329]	413K	1 (Push)	NA	RGB, L	xArm	Teleop
RH20T [335]	110K	42	147	RGB-D, L, F, A	4 robots	Teleop
RT-1 [16]	130K	12	700+	RGB, L	Google EDR	Teleop
OXE [17]	1.4M	527	160,266	RGB-D, L	22 robots	Mixed
DROID [330]	76K	86	NA	RGB-D, L	Franka	Teleop
FuSe [198]	27K	2	3	RGB, L, T, A,	WidowX 250	Teleop
RoboMIND [333]	107K	38	479	RGB-D, L	4 robots	Teleop
AgiBot World [94]	1M	87	217	RGB-D, L	AgiBot G1	Teleop

💭 Thought Inspiration

> Tips: What aspects did you question, how do you think it can be improved?

📜 Conclusion

> Tips: What was done, what problem was solved, innovations and shortcomings?